TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

Machine Learning & Artificial Intelligence
Internship Project Report

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Internship Organization	Cloudcredits Technologies
Project	Titanic Survival Prediction Using Machine Learning

Executive Summary

This project presents an end-to-end machine learning solution for predicting passenger survival during the Titanic disaster. The objective was to develop a binary classification model capable of determining whether a passenger survived based on demographic and travel-related attributes such as passenger class, gender, age, family information, fare, and embarkation port.

The project followed the complete machine learning lifecycle, including problem definition, data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model development, model evaluation, hyperparameter tuning, and deployment. Three machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were trained and compared.

After evaluation and optimization, Logistic Regression was selected as the final model due to its balanced performance, achieving an accuracy of 81.01% and an F1-score of 76.06%. The final model was deployed through a Streamlit web application that enables users to input passenger details and receive survival predictions with probability scores.

1. Introduction

Machine learning has become one of the most important technologies for predictive analytics and decision-making. By learning patterns from historical data, machine learning models can make predictions on unseen observations.

The Titanic disaster is one of the most widely used datasets in machine learning education. The dataset contains information about passengers onboard the RMS Titanic, including demographic details and survival outcomes. Because the actual survival status is known, the dataset is suitable for supervised learning classification tasks.

This project aims to develop a machine learning model capable of predicting passenger survival based on available passenger attributes while demonstrating the complete workflow of a practical machine learning solution.

2. Problem Definition

The objective of this project is to predict whether a passenger survived the Titanic disaster.

Problem Type

Binary Classification

Target Variable

Value	Meaning
0	Did Not Survive
1	Survived

Input Features

- Passenger Class (Pclass)
- Gender (Sex)
- Age
- Number of Siblings/Spouses (SibSp)
- Number of Parents/Children (Parch)
- Fare
- Embarked Port

The trained model returns a predicted survival outcome along with survival probabilities.

3. Dataset Description and Preparation

The project uses the Titanic dataset distributed through Seaborn's example dataset repository.

Dataset Overview

Attribute	Value
Records	891
Features	15
Training Records	712
Testing Records	179
Target Variable	survived

Missing Values

Feature	Missing Values
Age	177
Embarked	2
Deck	688
Embark Town	2

Data Cleaning

The preprocessing phase included:

- Missing value treatment
- Duplicate value inspection
- Feature selection
- Data transformation
- Training and testing split

Features with excessive missing values, such as deck, were not used in the final model.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify relationships between passenger attributes and survival outcomes.

4.1 Survival Distribution

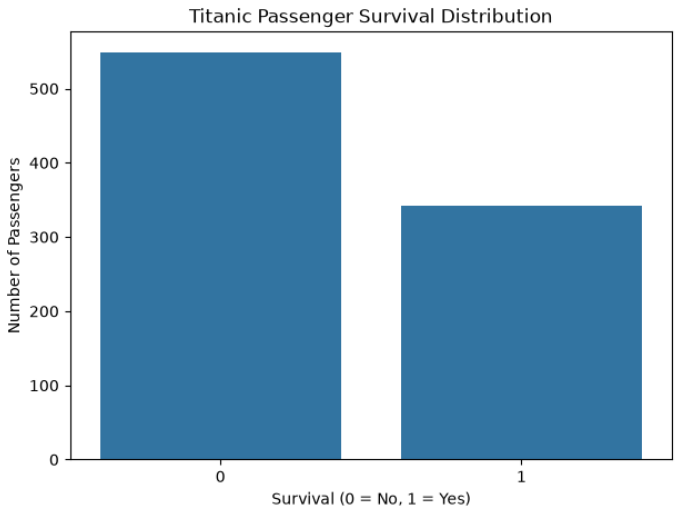


Figure 1: Titanic Passenger Survival Distribution

Observation

The dataset contains more non-survivors than survivors, indicating moderate class imbalance. Approximately 61.6% of passengers did not survive, while 38.4% survived.

4.2 Gender-Based Survival Analysis

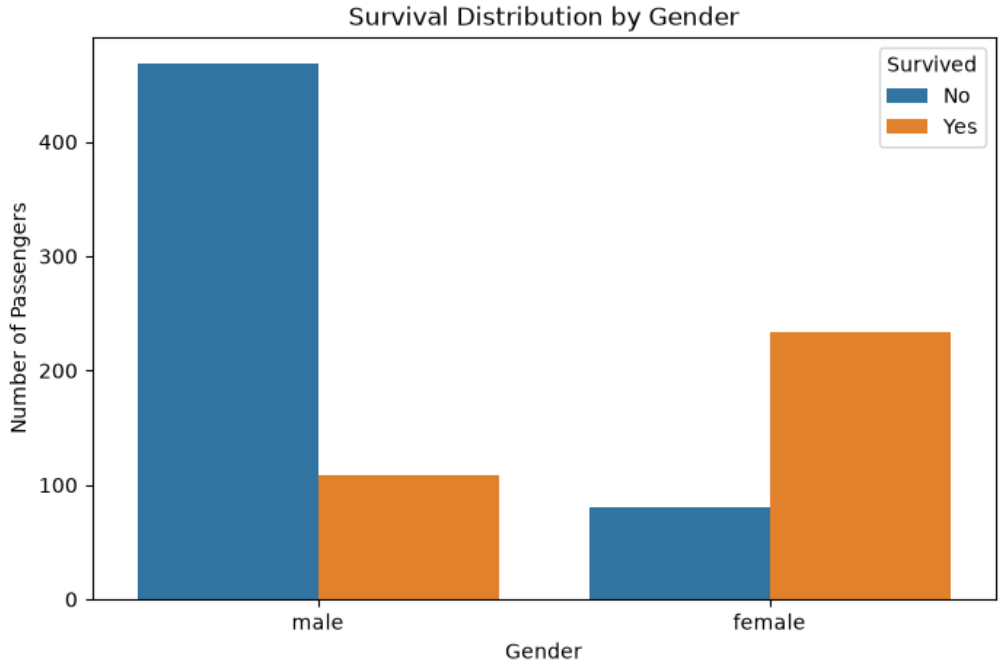


Figure 2: Survival Distribution by Gender

Observation

Female passengers had substantially higher survival rates than male passengers.

Gender	Survival Rate
Female	74.20%
Male	18.89%

Gender was identified as one of the most influential predictors of survival.

4.3 Passenger Class Analysis

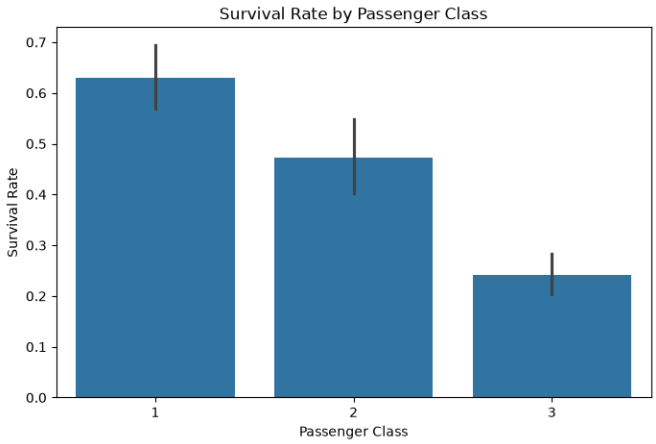


Figure 3: Survival Rate by Passenger Class

Observation

Passengers travelling in first class exhibited the highest survival rates, followed by second-class passengers. Third-class passengers experienced the lowest survival rates.

4.4 Age Analysis

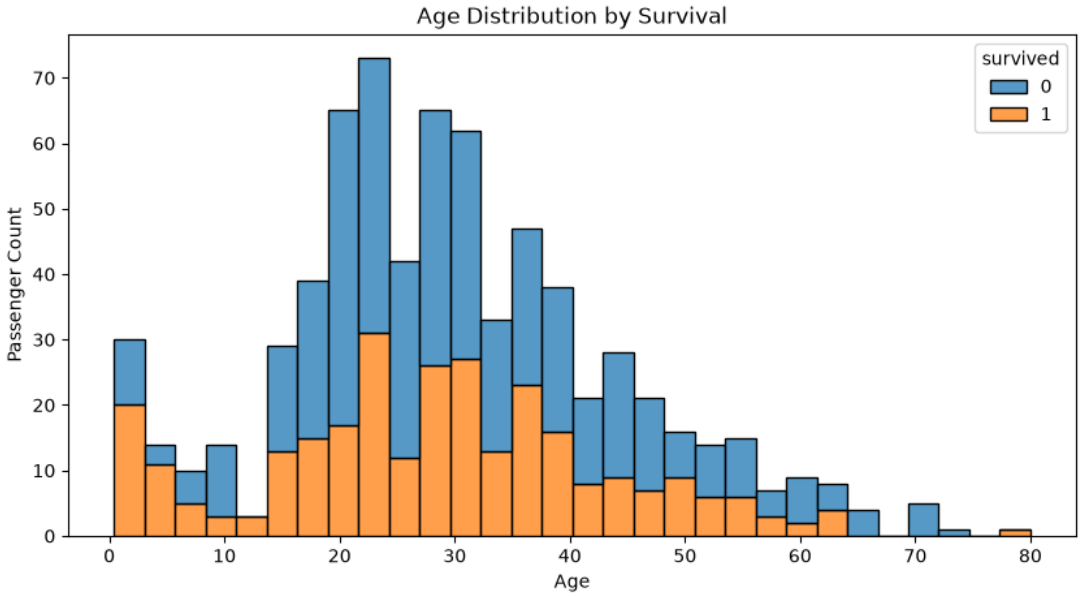


Figure 4: Age Distribution by Survival

Observation

Survival outcomes varied across different age groups. Younger passengers generally displayed better survival outcomes compared to older passengers.

4.5 Correlation Analysis

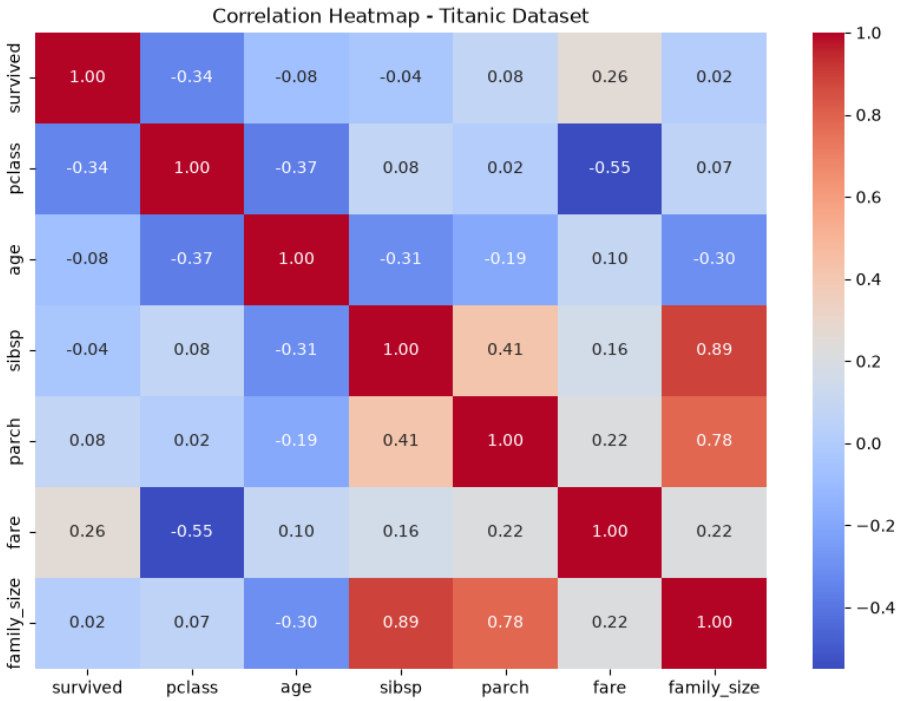


Figure 5: Correlation Heatmap

Observation

Passenger class demonstrated the strongest negative correlation with survival (-0.34), while fare exhibited a positive correlation (0.26). These variables were considered important predictors during model development.

5. Feature Engineering

Feature engineering was performed to investigate whether additional derived variables could improve predictive performance.

Engineered Features

Feature	Description
family_size	SibSp + Parch + 1
is_alone	Indicates if passenger travelled alone
fare_per_person	Fare divided by family size
age_group	Categorized age ranges

Although engineered features provided useful insights, the original feature set ultimately produced better performance and was retained for deployment.

6. Data Splitting and Preprocessing

The dataset was divided using an 80:20 training-testing ratio while preserving target class distribution through stratified sampling.

Numerical Features

- Age
- SibSp
- Parch
- Fare

Processing:

- Median Imputation
- StandardScaler

Categorical Features

- Pclass
- Sex
- Embarked

Processing:

- Most Frequent Imputation
- One-Hot Encoding

Scikit-Learn Pipelines and ColumnTransformer were used to ensure consistent preprocessing during training and deployment.

7. Model Development

Three machine learning algorithms were selected:

1. Logistic Regression
2. Decision Tree
3. Random Forest

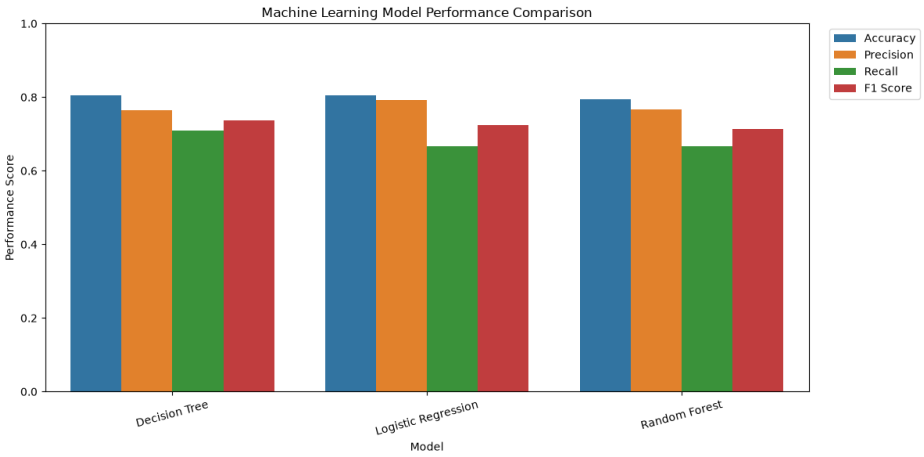


Figure 6: Baseline Model Performance Comparison

Baseline Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80.45%	79.31%	66.67%	72.44%
Decision Tree	80.45%	76.56%	71.01%	73.68%
Random Forest	79.33%	76.67%	66.67%	71.32%

8. Model Evaluation

Confusion matrices were used to analyze model predictions.

Logistic Regression

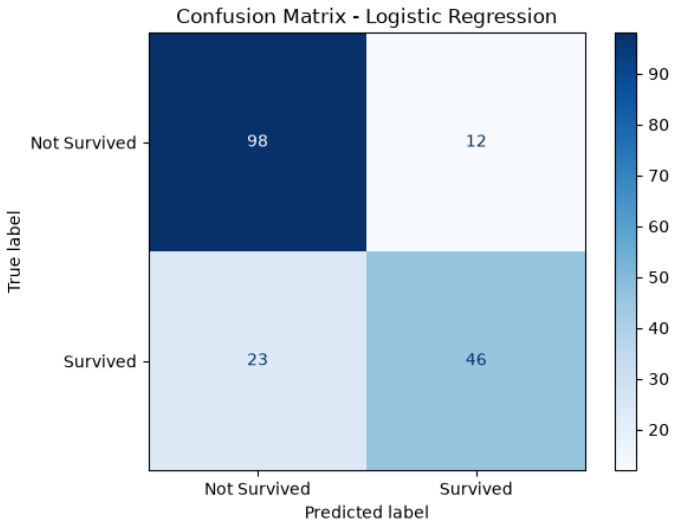


Figure 7: Logistic Regression Confusion Matrix

Decision Tree

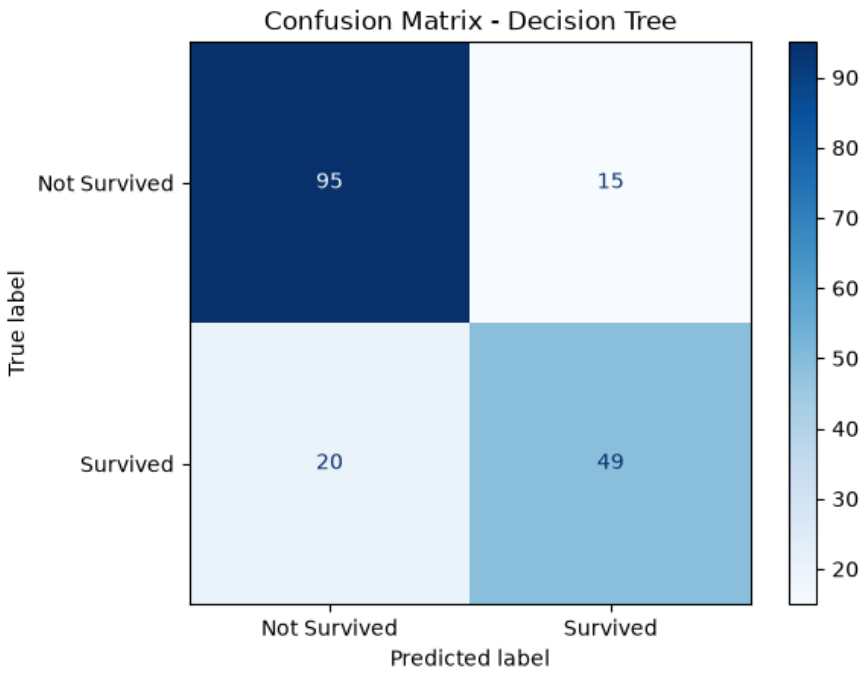


Figure 8: Decision Tree Confusion Matrix

Random Forest

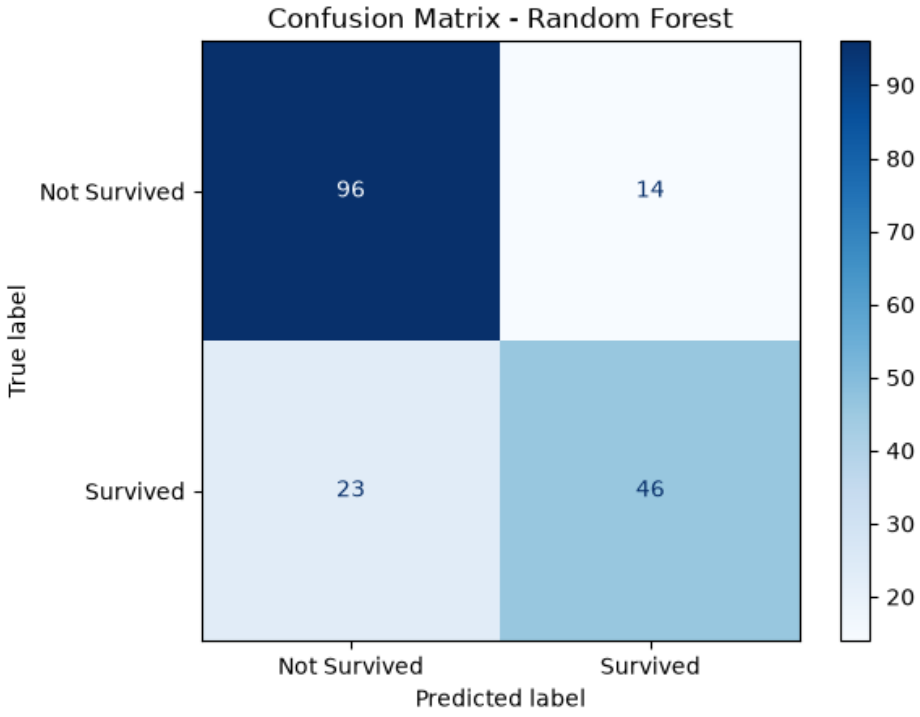


Figure 9: Random Forest Confusion Matrix

Evaluation Summary

All three models achieved competitive performance. Logistic Regression demonstrated the most balanced combination of precision, recall, and F1-score, making it a strong candidate for further optimization.

9. Model Optimization and Final Evaluation

Hyperparameter tuning was performed using RandomizedSearchCV with five-fold cross-validation.

Tuned Model Comparison

	Model	Accuracy	Precision	Recall	F1 Score
0	Tuned Logistic Regression	0.810056	0.739726	0.782609	0.760563
2	Tuned Random Forest	0.798883	0.746269	0.724638	0.735294
1	Tuned Decision Tree	0.776536	0.737705	0.652174	0.692308

Figure 10: Tuned Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	81.01%	73.97%	78.26%	76.06%
Random Forest	79.89%	74.63%	72.46%	73.53%
Decision Tree	77.65%	73.77%	65.22%	69.23%

Final Classification Report

```
Final Accuracy: 0.8100558659217877
Final Classification Report:

              precision    recall  f1-score   support

Not Survived       0.86      0.83      0.84       110
    Survived       0.74      0.78      0.76        69

    accuracy                           0.81       179
   macro avg       0.80      0.80      0.80       179
weighted avg       0.81      0.81      0.81       179
```

Figure 11: Classification Report

Final Metrics

- Accuracy: 81.01%
- Precision: 73.97%
- Recall: 78.26%
- F1 Score: 76.06%

Final Confusion Matrix

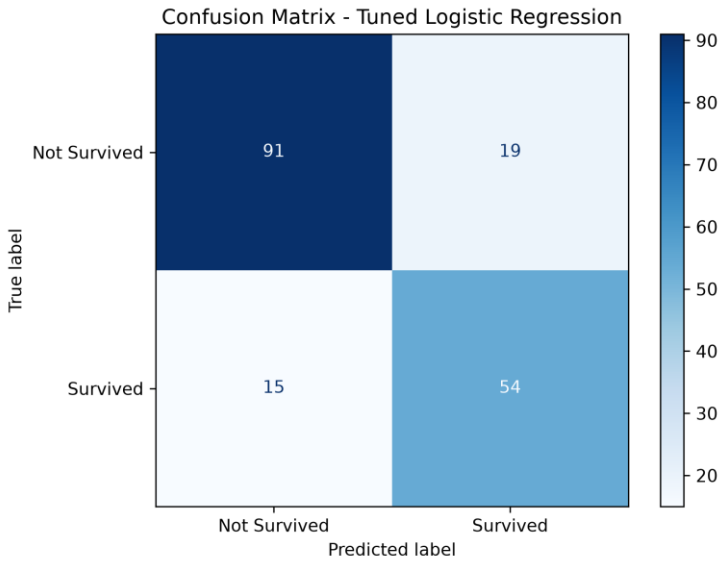


Figure 12: Tuned Logistic Regression Confusion Matrix

Interpretation

The final model correctly classified:

- 91 non-survivors
- 54 survivors

The model achieved a balanced trade-off between precision and recall.

10. Feature Importance Analysis

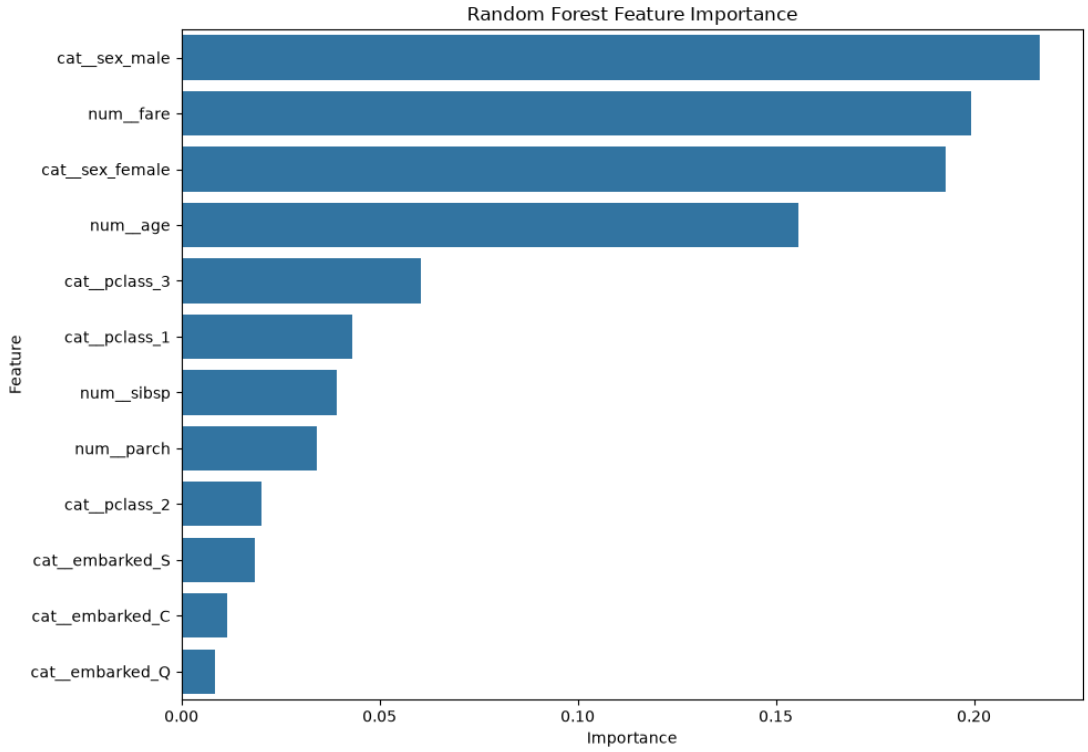


Figure 13: Random Forest Feature Importance

Observation

Feature importance analysis identified:

- Gender-related variables
- Fare
- Age

as the most influential factors affecting model predictions.

Feature importance reflects model behavior and should not be interpreted as evidence of causation.

11. Streamlit Application Deployment

The final Logistic Regression model was deployed using Streamlit to provide an interactive prediction interface.

Application Overview

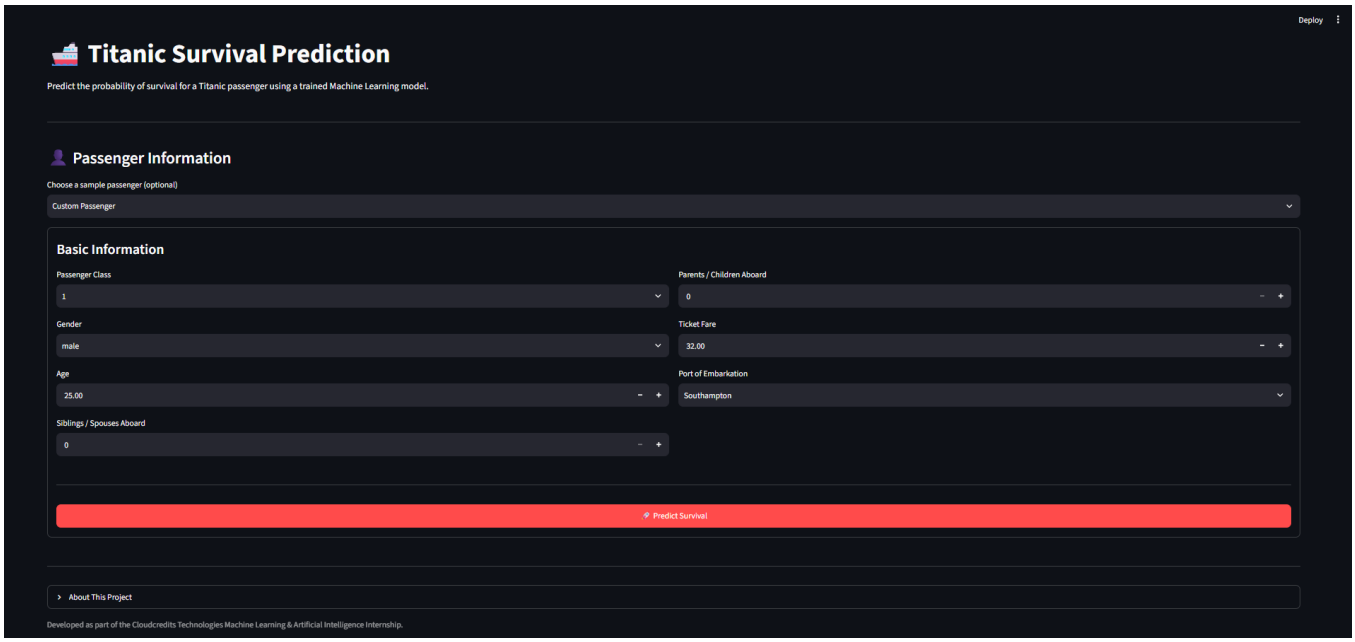


Figure 14: Streamlit Home Page

The home page provides a brief project overview and allows users to navigate the prediction workflow.

Passenger Input Form

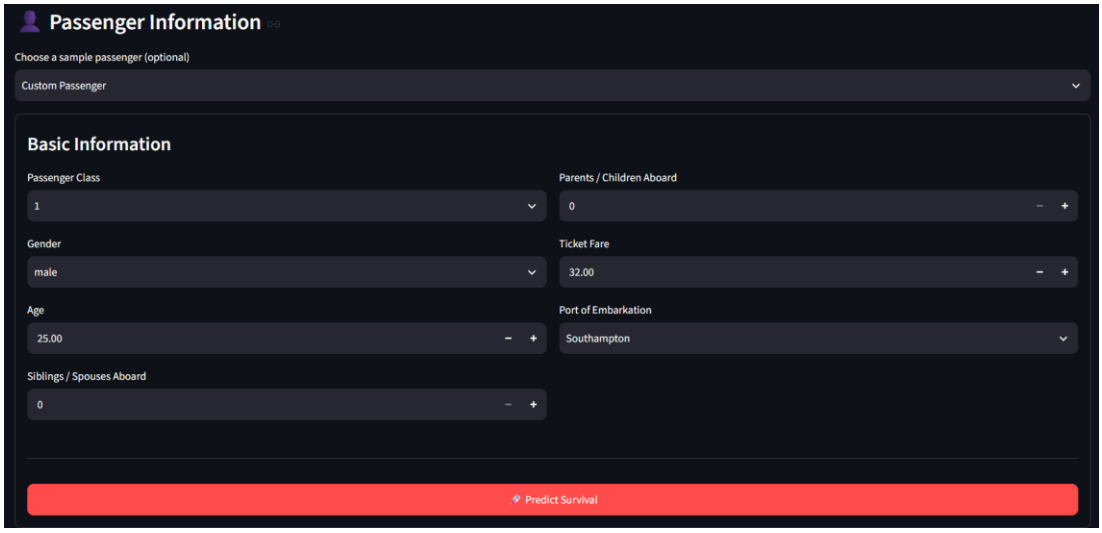


Figure 15: Passenger Input Form

Users can provide passenger information such as:

- Passenger Class
- Gender
- Age
- Fare
- Family Information
- Port of Embarkation

Prediction Results

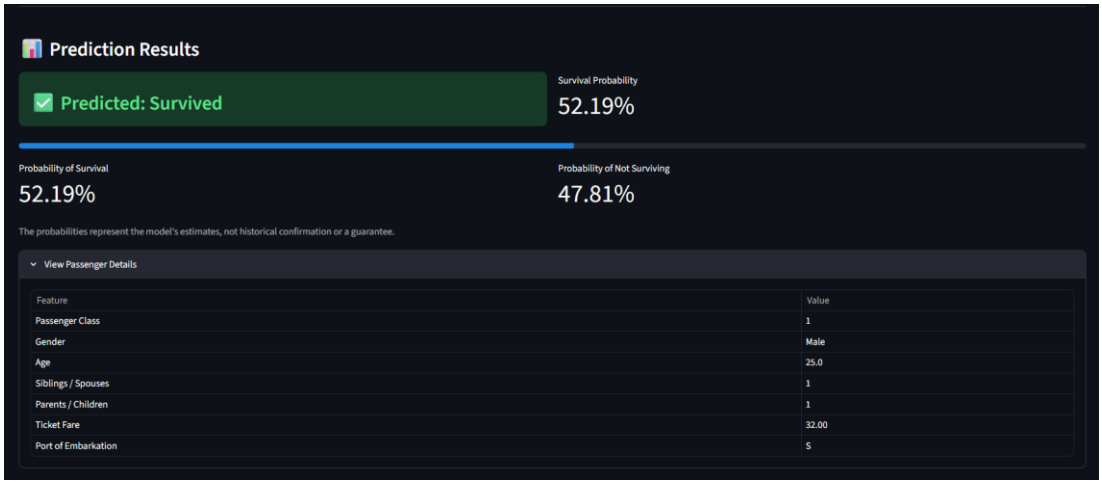


Figure 16: Prediction Results Screen

The application generates:

- Predicted Survival Outcome
- Probability of Survival
- Probability of Non-Survival

Example prediction:

Metric	Value
Predicted Class	Survived
Survival Probability	52.19%
Non-Survival Probability	47.81%

12. Project Structure

Path	Purpose
data/titanic.csv	Dataset
notebooks/01_Titanic_EDA.ipynb	EDA Notebook
src/preprocessing.py	Data Preprocessing
src/train_model.py	Model Training
src/predict.py	Prediction Utility
models/titanic_survival_model_retained.pkl	Saved Model
app.py	Streamlit Application
test_prediction.py	Testing Script
images/	Visual Assets

13. Limitations and Future Scope

Limitations

- Historical dataset with limited size.
- Results may not generalize to modern scenarios.
- Model learns associations rather than causation.
- Performance depends on available passenger attributes.

Future Scope

- Evaluate additional algorithms.
- Implement ensemble learning approaches.
- Perform probability calibration.
- Train using larger datasets.
- Deploy using cloud infrastructure.

14. Conclusion

This project successfully implemented a complete machine learning workflow for predicting Titanic passenger survival. The project involved data preprocessing, exploratory data analysis, feature engineering, model comparison, hyperparameter tuning, evaluation, and deployment.

Among all evaluated models, Tuned Logistic Regression achieved the best overall performance with 81.01% accuracy and a 76.06% F1-score. The deployment phase demonstrated practical application of machine learning concepts through a Streamlit-based prediction system.

The project follows the prescribed machine learning workflow and demonstrates the practical implementation of classification techniques in a real-world predictive analytics problem.

Organization: Cloudcredits Technologies

Project: Titanic Survival Prediction Using Machine Learning